Computer Speech and Language 90 (2025) 101752

Contents lists available at ScienceDirect

Computer Speech & Language

journal homepage: www.elsevier.com/locate/csl

Modeling correlated causal-effect structure with a hypergraph for document-level event causality identification☆

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ARTICLE INFO

Keywords:
Event causality identification
Correlated cause–effect structure
Hypergraph convolutional network
Pre-trained language model

ABSTRACT

Document-level event causality identification (ECI) aims to detect causal relations in between event mentions in a document. Existing approaches for document-level event causality identification detect the causal relation for each pair of event mentions independently, while ignoring latent correlated cause–effect structure in a document, i.e., one cause (effect) with multiple effects (causes). We argue that identifying the causal relation of one event pair may facilitate the causality identification for other event pairs. In light of such considerations, we propose to model the correlated causal-effect structure by a hypergraph and jointly identify multiple causal relations with the same cause (effect). In particular, we propose an event-hypergraph neural encoding model, called EHNEM, for document-level event causality identification. In EHNEM, we first initialize event mentions' embeddings via a pre-trained language model and obtain potential causal relation of each event pair via a multilayer perceptron. To capture causal correlations, we construct a hypergraph by integrating potential causal relations for the same event as a hyperedge. On the constructed event-hypergraph, we use a hypergraph convolutional network to learn the representation of each event node for final causality identification. Experiments on both EventStoryLine corpus and English-MECI corpus show that our EHNEM model significantly outperforms the state-of-the-art algorithms.

1. Introduction

Document-level event causality identification (ECI) aims to detect whether there exists a causal relation between two event mentions in a document. It is of great importance for many semantic web applications, such as knowledge graph construction (Heindorf et al., 2020; Khatib et al., 2020), information extraction (Yang et al., 2022; Xiang and Wang, 2019),question answer (Bondarenko et al., 2022; Sui et al., 2022), knowledge reasoning (Yao et al., 2021; Feder et al., 2022), and etc. Fig. 1 illustrates an example of annotated event mentions in a crime topic document with causality labels in the Event StoryLine Corpus (ESC) (Caselli and Vossen, 2017). Where event mentions are annotated as one or more words in raw sentence that most clearly expresses an event occurrence, and causal relation can exist in between both intra-sentence or cross-sentence event mentions. We note that an event may also contain some event arguments with a specific role, such as the time, place and participants of the event, to form a structured event. In this paper, we only use event mentions to identify causal relations between event pairs.

☆ Financially supported by self-determined research funds of CCNU from the colleges' basic research and operation of MOE (Grant No: CCNU24XJ029) and National Natural Science Foundation of China (Grant No: 62172167).

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https://doi.org/10.1016/j.csl.2024.101752
Received 31 July 2023; Received in revised form 13 August 2024; Accepted 8 November 2024

Available online 19 November 2024
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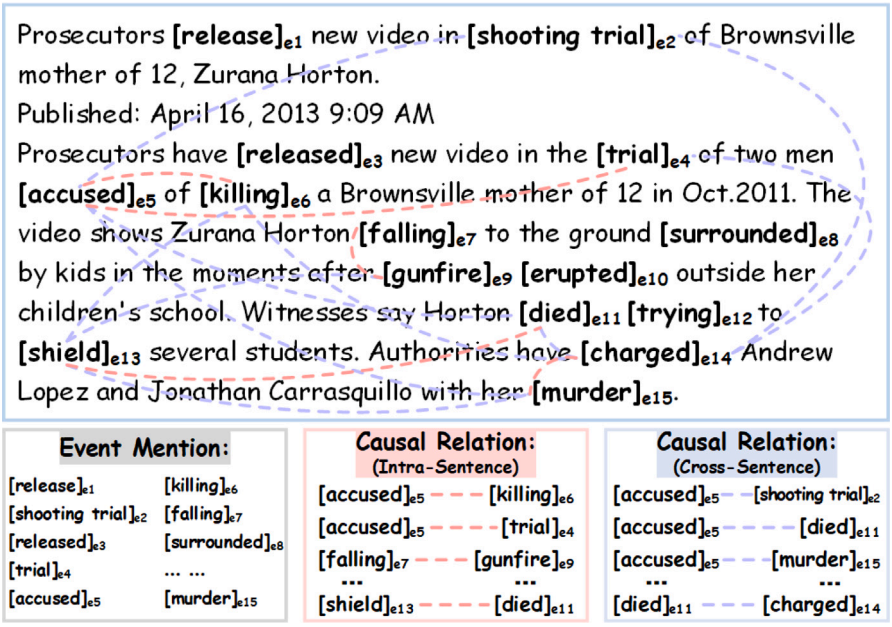


Fig. 1. Illustration of annotated event mentions and causality labels for a crime topic document in the ESC 0.9 corpus.

Some methods have employed external knowledge bases to obtain causal knowledge for augmenting causality identification. For example, the FrameNet (Baker et al., 1998) contains abundant structured triples of events, entities, and relations. Both (Zuo et al., 2020) and (Liu et al., 2020) use such knowledge triplets in the FrameNet to boost representation learning. Moreover, the ConceptNet, a graph-structured knowledge base (Speer et al., 2017), as well as the WordNet (Miller, 1995) and VerbNet (Schuler, 2006), each a lexical knowledge base have also been used to obtain external causal knowledge for the ECI task (Cao et al., 2021; Zuo et al., 2021a).

Although these external knowledge bases can effectively augment causality detection, they are mainly used for identifying event causality in the sentence-level, but cannot fully utilize document-level contextual semantics for cross-sentence causality identification. Some recent methods design sophisticated neural models to learn a kind of document-level semantic representation for each event, such as the Rich Graph Convolutional Network (RichGCN) (Phu and Nguyen, 2021), Event Relational Graph Transformer (ERGO) (Chen et al., 2022), and Graph-based Event Structure Induction model (GESI) (Fan et al., 2022).

However, all these graph-based neural models have ignored the latent correlated cause–effect structure in a document, i.e., one cause (effect) with multiple effects (causes), such as the example illustrated in Fig. 1. We argue that the events' causal relations in a document are often interdependent and identifying one cause–effect pair may facilitate the identification of other causes (effects) with the same effect (cause). For example, the effect event $[accused]_{e5}$ has multiple cause events including $[shootingtrial]_{e2}$, $[killing]_{e6}$, $[died]_{e11}$, and etc.; While identifying the causal relation between $[killing]_{e6}$ and $[accused]_{e5}$ may facilitate the identification of causal event pair ($[accused]_{e5}$, $[murder]_{e15}$), ($[accused]_{e5}$, $[trial]_{e4}$), and ($[accused]_{e5}$, $[died]_{e11}$), that have a same cause (effect) event $[accused]_{e5}$.

Besides, we argue that the identification of causal event pair with co-reference events can also benefit from such document correlated causal-effect structure learning. For example, the co-reference event $[trial]_{e4}$ and $[shooting\ trial]_{e2}$ both have causal relation with event $[accused]_{e5}$, as illustrated in Fig. 1; While the cross-sentence causal relation between $[accused]_{e5}$ and $[shooting\ trial]_{e2}$ is more difficult to identify than the identification of intra-sentence event causality $[accused]_{e5}$ and $[trial]_{e4}$ for its discontinuous semantics. As such, identifying the intra-sentence event causality may also facilitate the identification of cross-sentence event causality with co-reference events.

Motivated from such considerations, we propose to model latent correlated causal-effect structure at the document-level by an event-hypergraph for jointly identifying causal relations. In contrast to conventional graph neural network where one relation edge can only connect two event nodes, the hyperedge in a hypergraph can connect more than two event nodes (Feng et al., 2019). As a result, the representation of multiple event nodes can be learned correlatively based on such kind of hyperedge. To this end, we propose utilizing hyperedges in a hypergraph to model the latent correlated cause–effect structure in a document for event causality identification.

In this paper, we propose an event-hypergraph neural encoding model, called EHNEM, for document-level event causality identification. In EHNEM, we first use a pre-trained language model (PLM) to initialize event mentions' embeddings and obtain potential causal relation of each event-pair via a multilayer perceptron. To capture latent causal correlations, we construct a hypergraph by integrating potential causal relations for the same event as a hyperedge, in which each node represents an event and

a hyperedge represents potential causal relations in between its connected event nodes. On the constructed event-hypergraph, we use a hypergraph convolutional network to learn the representation of each event node for final causality identification. Experiment results have shown that our proposed EHNEM significantly outperforms the state-of-the-art algorithms in terms of much higher F1 score on both EventStoryLine 0.9 corpus and English-MECI corpus.[1].

Our main contributions are summarized as follows:

1. Propose an event-hypergraph neural encoding model (EHNEM) to model the correlated causal-effect structure by a hypergraph for document-level event causality identification.
2. Propose an event semantic encoding module to initialize events' embedding and detect potential causal relations of each event-pair.
3. Propose an event-hypergraph learning module to model correlated causal-effect structure and jointly identify multiple causal relations.
4. Experiment results on both EventStoryLine 0.9 corpus and English-MECI corpus validate the effectiveness of our proposed EHNEM.

The rest of this article is organized as follows: Section 2 reviews some approaches of knowledge-based, reasoning-based and graph-based event causality identification. Section 3 introduces our EHNEM model including event semantics encoding, event-hypergraph learning and causality classification module. We present the experiments settings in Section 4 and compare the performance of our EHNEM model with some discussion in Section 5. Finally, Section 6 concludes this article and discusses our future work.

2. Related work

2.1. Knowledge-based causality identification

Existing knowledge bases, like the FrameNet (Baker et al., 1998), ConceptNet (Speer et al., 2017), and etc., have stored large-scale structured information that can provide some kind of abstract causal knowledge for causality reasoning. Some studies try to learn some causal patterns from external knowledge bases to implement a kind of knowledgeable event causality inference (Cao et al., 2021; Liu et al., 2020). For example, Cao et al. (2021) perform event causality reasoning based on the induced descriptive knowledge and relational path knowledge from the graph-structured knowledge base ConceptNet. Liu et al. (2020) enhance the ability of their neural model for previously unseen causal relation identification by mining some event-agnostic and context-specific patterns from the ConceptNet. Zuo et al. (2021a) learn some context-specific causal patterns from external causal statements based on a contrastive transfer learning framework.

Besides, some studies directly use the large amount of elaborately created items in lexical knowledge bases for data expansion and model training augmentation (Zuo et al., 2020, 2021b). For example, Zuo et al. (2020) distantly label event causality sentence based on the synonyms of each annotated causal event pair extracted from two lexical knowledge bases WordNet (Miller, 1995) and VerbNet (Schuler, 2006). They further generate new causal sentences interactively for event causality data augmentation based on existing knowledge bases (Zuo et al., 2021b). These approaches, however, ignore the contextual information and semantic evidence in a document for event causality identification.

2.2. Reasoning-based causality identification

Some studies try to identify event causalities by relation reasoning like event relation chain reasoning (Yuan et al., 2023), event counterfactual reasoning (Mu and Li, 2023), and etc. For example, Yuan et al. (2023) design various types of attention mechanism to capture long-distance semantic dependence for an event mention in document and learn a kind of sparse event representation as intra-sentential features. They further constitute a possible reasoning chain based on intra-sentential event pairs for inter-sentential event causality reasoning. Mu and Li (2023) propose a counterfactual reasoning mechanism to estimate context-keywords bias and event-pairs bias and then subtract the biases from factual predictions for event causality identification. Zhang et al. (2024) propose a multi-task learning framework to enhance event causality identification with rationale and structure-aware causal question answering.

The recent pre-train, prompt, and predict paradigm models the probability of text in Pre-trained Language Models(PLMs) to perform a prediction task, which has achieved promising results for many NLP tasks (Seoh et al., 2021; Xiang et al., 2022) and has been successfully applied in the ECI task (Shen et al., 2022; Liu et al., 2020). For example, Liu et al. (2020) use an event mention masking generalization mechanism to encode some event causality patterns for causal relation reasoning. Shen et al. (2022) use a masked language model to predict a causality answer word between two event mentions for causal relation classification, assisted by two derivative prompt tasks to identify the explicit causality based on the causal cue word detection. Man et al. (2024) leverage an optimal transport to automatically select the most important sentences and words from full documents for input into the generative language model T5, which is trained to generate causal relations.

[1] Source codes will be released later.

2.3. Graph-based causality identification

The graph-based approaches usually model the ECI task as a node classification problem and apply graph neural networks for learning event node representations from document-level contextual semantics (Phu and Nguyen, 2021; Cao et al., 2021; Fan et al., 2022; Pu et al., 2023; Hu et al., 2023; He et al., 2024). For example, Phu and Nguyen (2021) model diverse connections in between words of a document, like positional connection, syntactic dependency and etc., for the graph construction. They use a graph convolutional network to learn the event mention nodes' representations, and identify causalities through event node pair classification. Fan et al. (2022) build the explicit connections between events and their mentions, as well as their contexts, to construct a co-occurrence graph for event causal relation identification. Pu et al. (2023) predict causality-related events as a kind of event causal label information and construct an event pair interaction graph to learn the representation of each event pair node. Hu et al. (2023) utilize a graph neural network (GNN) based event aggregator to integrate event-centric structure information, and a long short term memory (LSTM) based path aggregator to capture event-associated structure information between two events for causal relation classification. He et al. (2024) introduce a heterogeneous graph interaction network to model the long-distance dependencies between events and propose a multi-granularity contrastive transfer learning module to align the causal representations across languages.

Besides node classification, some studies also infer the potential causal edges in an event graph for causal relation identification (Zhao et al., 2021; Chen et al., 2022; Tao et al., 2023; Chen et al., 2023; Liu et al., 2024). For example, Zhao et al. (2021) initialize event nodes' embeddings from a document-level encoder and use a graph inference mechanism for causal edge prediction. Chen et al. (2022) build an event relational graph where each node denotes a pair of events. They further use a graph transformer model to capture potential causal chains among nodes. However, all these graph-based methods have not jointly considered pairwise and document-wise event causality. Tao et al. (2023) generate event causality graph based on a pre-trained language model and perform structural discriminative training to further improve the understanding of event causality. Chen et al. (2023) predict and encode event centrality information into the constructed heterogeneous event interaction graph for causality reasoning. Liu et al. (2024) design an iterative learning and identifying framework that first construct an event causality graph, on which events' causal structure representations are updated for boosting causal identification.

The proposed EHNEM constructs a hypergraph to learn the causal correlations structure for event causality identification.

3. Method

Fig. 2 illustrates our EHNEM model, including the event semantics encoding, event-hypergraph construction and learning, and document-level causality identification. The event semantics encoding uses a pre-trained language model to encode event semantics as well as predict potential causal relations of each event-pair for next correlated causal-effect structure learning. Then we construct a document-level causal hypergraph based on predicted causal relations to fine-tune events' representations by a hypergraph convolutional neural network. Finally, two kinds of representations are concatenated for document-level event causality identification.

3.1. Event semantics encoding

The event semantic encoding (ESE) module is to initialize events' representations and detect a *potential* causal relation for each event-pair. The ESE module is built upon a pre-trained language model (PLM), here the RoBERTa (Liu et al., 2019), to encode an initial representation for each event mention Evt_i, as well as predicting a potential causal relation Rel_k for each event-pair. The RoBERTa is an advanced PLM proposed by Facebook, which is pre-trained using a *cloze task* on a large amount of unlabeled text in an unsupervised way.

As the event mentions are annotated by a few words in a sentence, we use the raw sentence of event mentions S_i and S_j for each event pair $x_k = (Evt_i, Evt_j)$ to include the full contextual semantics, as the input of RoBERTa. Let $S_i = [v_i^1, v_i^2, \dots, v_i^m]$ and $S_j = [v_j^1, v_j^2, \dots, v_j^n]$ denote raw sentence containing the annotated event mention Evt_i and Evt_j, where $Evt_i \subset S_i$, $Evt_j \subset S_j$, where v_is and v_js are all vocabulary tokens. Note that, the two events of an event pair may also be in the same sentence, i.e. $S_i = S_j$, as such the input sentence T of RoBERTa is denoted as:

$$T(x_k) = [t_k^1, t_k^2, \dots, t_k^L] = \begin{cases} [v_i^1, v_i^2, \dots, v_i^m, v_j^1, v_j^2, \dots, v_j^n], & if \quad S_i \neq S_j, \\ [v^1, v^2, \dots, v^L], & if \quad S_i = S_j, \end{cases} \tag{1}$$

where, L is the length of input sentence.

After RoBERTa encoding, we obtain a hidden state for each token of the input sentence:

$$\mathbf{H}_k = [\mathbf{h}_k^1, \mathbf{h}_k^2, \dots, \mathbf{h}_k^L] = RoBERTa([t_k^1, t_k^2, \dots, t_k^L]), \tag{2}$$

where $\mathbf{H}_k \in \mathbb{R}^{L*d}$ and d is the dimension of hidden state. The event mentions for kth event pair are initially represented by an average pooling of their contained tokens' hidden state $\mathbf{e}_k^i = 1/l \sum \mathbf{h}_k^i$ and $\mathbf{e}_k^j = 1/l \sum \mathbf{h}_k^j$, where $\mathbf{e}_k^j, \mathbf{e}_k^j \in \mathbb{R}^d$. $\mathbf{h}_k^i$, $\mathbf{h}_k^j$ are the hidden states of event mention contained tokens that are subsets of input sentence H_k and l is the length of event mention contained tokens.

To detect the existence of a potential causal relation for each event-pair, we use a Multilayer Perceptron (MLP) to transform the concatenation of two event mention vectors to a causality probability. The transition function is:

$$P_k = ReLU(\mathbf{W}_k[\mathbf{e}_k^i \parallel \mathbf{e}_k^j] + \mathbf{b}_k), \tag{3}$$

where $\parallel$ stands for the concatenation operation and $\mathbf{W}_k$ and $\mathbf{b}_k$ are learnable parameters.

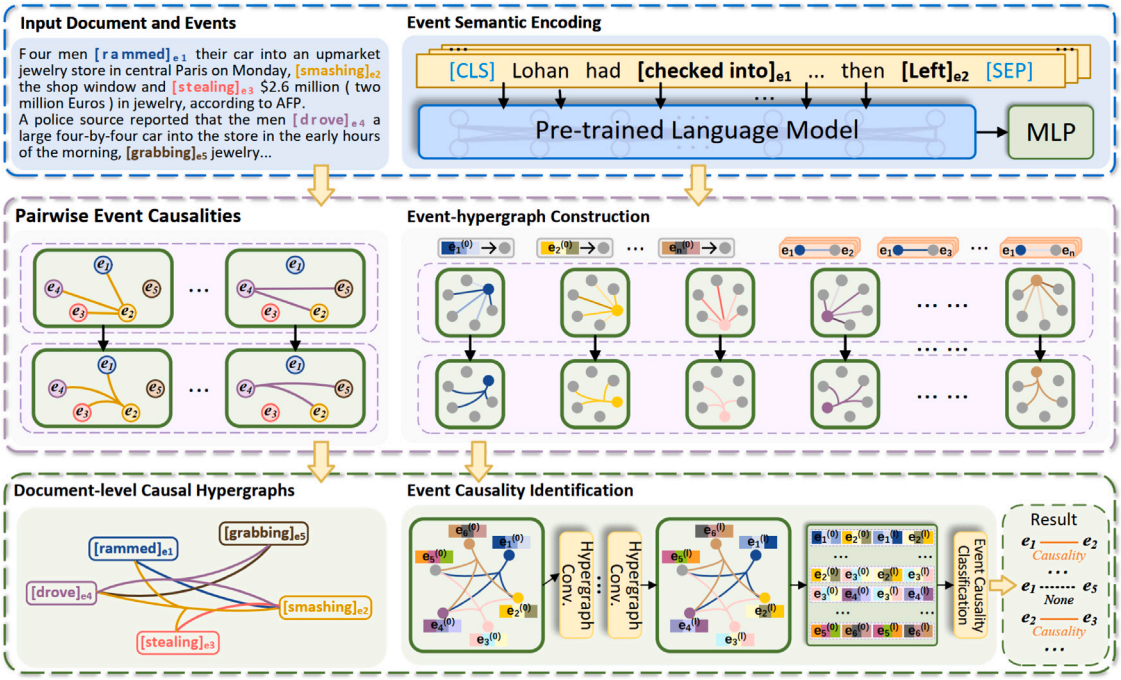


Fig. 2. Illustration of our EHNEM model.

We pre-train the MLP and fine-tune the RoBERTa by the ground truth causal relations in the training dataset. The prediction with the highest overall F1 in the validation dataset is adopt as potential causal relations Rel_k for the event-hypergraph construction. The cross entropy loss is adopted as cost function.

3.2. Event-hypergraph learning

This module is to model correlated causal relations among events via a hypergraph and fine-tuning events' representations in the document-level.

3.2.1. Construction of event-hypergraph

We first construct an event-hypergraph $\mathcal{G} = (\mathcal{E}, \mathcal{R})$ for a document, where $\mathcal{E}$ is the event node set and $\mathcal{R}$ the hyperedge set. As an event mention is differently encoded by the RoBERTa in each event-pair after the ESE module, we use the average of their vectors $\mathbf{e}_n$ as the initial hypergraph node representation $\mathbf{E}^{(0)} = (\mathbf{e}_1^0, \dots, \mathbf{e}_n^0)$, where $\mathbf{e}_n \in \mathbb{R}^d$ and $\mathbf{E}^{(0)} \in \mathbb{R}^{n \times d}$. The hyperedges are used to model the possibly correlated causal-effect structure of different events. According to the ESE output, an event node may be with multiple potential causal relations to several other nodes, which are used to create a hyperedge.

We construct the hypergraph incidence matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{R}|}$ with node as row and hyperedges as column. As illustrated in Fig. 2, each hyperedge has one vertex as the center node and connects several other nodes by potential causal relations Rel_k. Let $R = (r_1, \dots, r_m)$ denotes the hyperedge set, r_m the hyperedge of mth vertex, and $\tilde{\mathbf{E}}_m$ denotes the node set of connected events to the mth vertex. The element of hypergraph incidence matrix $\mathbf{A}(e_n, r_m)$ can be formulated as:

$$\mathbf{A}(e_n, r_m) = \begin{cases} 1, & if \quad e_n \in \tilde{\mathbf{E}}_m, \\ 0, & if \quad e_n \notin \tilde{\mathbf{E}}_m, \end{cases} \tag{4}$$

where e_n denotes the nth event node in document, if an event e_n is a node to the hyperedge r_m on the graph $\mathcal{G}$, $\mathbf{A}(e_n, r_m) = 1$; Otherwise, $\mathbf{A}(e_n, r_m) = 0$.

Note that in the training phase, we use the ground truth relations for the event-hypergraph construction; While in the testing phase, we use the predicted ones by the ESE module.

3.2.2. Correlated causal-effect structure learning:

On the event-hypergraph, we adopt a hypergraph convolution network (HGNN) (Feng et al., 2019) to fine-tune the document-level representation for each event node. The HGNN employs a kind of flexible hyperedges to connect multiple event nodes (usually

more than two) and construct a hypergraph for convolutional operations. This enables the HGNN to learn hidden layer representation considering the high-order data structure of the hypergraph. The transition function is as follows:

$$\mathbf{E}^{(l)} = \sigma(\mathbf{D}_e^{-\frac{1}{2}}\mathbf{A}\mathbf{I}\mathbf{D}_r^{-1}\mathbf{A}^{\top}\mathbf{D}_e^{-\frac{1}{2}}\mathbf{E}^{(l-1)}\Theta^{(l-1)}), \tag{5}$$

where σ is the ReLU activation function, $\mathbf{I}$ is an identity matrix, and $\Theta \in \mathbb{R}^{d\times d}$ the learnable parameters. $\mathbf{D}_e$ and $\mathbf{D}_r$ denote the diagonal matrices of the node degrees and edge degrees, respectively. They are used for normalization. For a vertex $e \in \mathcal{E}$ and a hyperedge $r \in \mathcal{R}$, their degrees are computed as follows:

$$d(e) = \sum_{r\in\mathcal{R}} w(r)a(e,r), \tag{6}$$

$$d(r) = \sum_{e\in\mathcal{E}} a(e,r), \tag{7}$$

where $a(e,r)$ is the element of incidence matrix $\mathbf{A}$ and $w(r)$ is an element on the diagonal of matrix $\mathbf{I}$ that indicates the weight of hyperedge r.

According to Eq. (5), the process of hypergraph convolution layer is as follows:

- **Step 1:** The learnable parameter matrix at $l-1$ layer $\Theta^{(l-1)}$ is used to transform the $l-1$ layer node features $\mathbf{E}^{(l-1)}$ to the new node features. Note that the initial node features $\mathbf{E}^{(0)}$ is obtained from the event semantic encoding module;
- **Step 2:** The new node features on the hyperedges are gathered to obtain the hyperedge features by the multiplication of $\mathbf{A}^{\top}$;
- **Step 3:** The related hyperedge features are associated to obtain the l layer node features $\mathbf{E}^{(l)}$, which is implemented by multiplying incidence matrix $\mathbf{A}$;

The outputs after l convolution layers include the document-level events' representation $\mathbf{E}^{(l+1)} \in \mathbb{R}^{n\times d}$, and the last layer outputs $\mathbf{E}^{(l)} = (\mathbf{e}_1^l, \ldots, \mathbf{e}_n^l)$ are used for the final causal relation classification.

3.3. Event causality classification

We argue that the initial event-hypergraph nodes' encodings from the ESE module output contain a kind of local contextual semantics for each event mention, which are also useful for final causal relation identification. To this end, we concatenate the representations learned from the event-hypergraph $\mathbf{E}^{(l)}$ with its initial representation $\mathbf{E}^{(0)}$ as the final representation for an event pair $\mathbf{v}_k \in \mathbb{R}^{4d}$ for final causal relation classification. The causality classifier is a fully connected layer with softmax to transform each event-pair representation $\mathbf{v}_k$ to a probability distribution $\hat{\mathbf{y}}$ for predicting the event causal relation:

$$\hat{\mathbf{y}} = softmax(\mathbf{W}_c\mathbf{v}_k + \mathbf{b}_c), \tag{8}$$

where $\mathbf{W}_c$, $\mathbf{b}_c$ are learnable parameters.

The EHNEM model is trained at the document-level. We compute the cross entropy loss for each event-pair in a document and their summation is used as the cost function:

$$J(\theta) = -\frac{1}{K}\sum_{k=1}^{K}\mathbf{y}^{(k)}\log(\hat{\mathbf{y}}^{(k)}) + \lambda\|\theta\|^2, \tag{9}$$

where $\mathbf{y}^{(k)}$ and $\hat{\mathbf{y}}^{(k)}$ are the gold label and predicted label of the kth event-pair, respectively, and λ and θ are the regularization hyper-parameters.

4. Experiment settings

This section presents our experiment settings, including the EventStoryLine and MECI dataset, competitors, and parameter settings.

4.1. Datasets

Our experiments are conducted on two widely used document-level event causality identification datasets, that is, the EventStoryLine 0.9 Corpus (ESC) (Caselli and Vossen, 2017) and the English dataset in Multilingual Event Causality Identification (English-MECI) Corpus (Lai et al., 2022).

EventStoryLine[2] is a widely used document-level event causality dataset for the ECI task. The ESC dataset contains 22 topics, 258 documents and in total 5334 event mentions from various news web-sites. A total number of 5625 event pairs are annotated with causal relations (1770 intra-sentence and 3855 cross-sentence). Following the standard data splitting (Gao et al., 2019), we use the last two topics as the development set, and conduct 5-fold cross-validation on the remaining 20 topics. The average results of precision (P), recall (R), and $F1$ score are adopted as performance metrics.

[2] https://github.com/tommasoc80/EventStoryLine

MECI[3] is a multi-language event causality identification dataset, and we conduct our experiments on English language dataset (English-MECI), which contains 438 documents from 5 topics on Wikipedia. There are in total 8732 event mentions in the English-MECI corpus. A total number of 2050 event pairs are annotated with causal relations, among which 1,889 event pairs are from intra-sentence event pairs and 161 causal relations are from cross-sentence event pairs. Following the standard data splitting provided by Lai et al. (2022), 263 documents are used as the training set, 88 as the development set and 87 as the testing set. We randomly select 5 different initialization seeds and averaged the results of precision (P), recall (R), and F1 score in our experiments.

4.2. Competitors

We compare with the following competitors:

- ILP (Gao et al., 2019) uses integer linear programming to identify causal relations based on document-level causal structures constraints.
- RichGCN (Phu and Nguyen, 2021) uses a graph convolutional network to learn a document context-augmented representation of event-pairs.
- GESI (Fan et al., 2022) builds an event co-reference graph to identify event causal relation by a graph convolutional network.
- ERGO (Chen et al., 2022) builds an event relational graph using an event-pair as a node to capture causation transitivity via a transformer-like neural network.
- CHEER (Chen et al., 2023) builds a heterogeneous event interaction graph to integrate event centrality information for causality reasoning.
- SENDIR (Yuan et al., 2023) constitutes a reasoning chain based on intra-sentential event pairs for inter-sentential event causality reasoning.
- KnowDis (Zuo et al., 2020) uses external lexicon knowledge to augment event causality data.
- KnowMMR (Liu et al., 2020) uses external knowledge to mine some event causality patterns.
- CauSeRL (Zuo et al., 2021a) adopts a contrastive strategy to transfer learned external causal statements.
- LearnDA (Zuo et al., 2021b) leverages existing knowledge bases to interactively generate training data.
- LSIN (Cao et al., 2021) uses a graph induction model to learn external structural and relational knowledge.
- DPJL (Shen et al., 2022) leverages two derivative prompt tasks to identify the explicit and implicit causality.
- CF-ECI (Mu and Li, 2023) estimates context-keywords bias and event-pairs bias to identify causal relations by counterfactual reasoning.
- ECLEP (Pu et al., 2023) integrates event causality-related information into event pair interaction graph.

We note that not all of these competitors are designed for document-level causal relation identification, and some are only applicable for intra-sentence event-pair causality identification.

4.3. Parameter settings

We use the pre-trained RoBERTa (Liu et al., 2019) model with 768-dimension base version provided by the HuggingFace transformers[4] (Wolf et al., 2020). The RoBERTa is an advanced PLM proposed by Facebook[5], which is pre-trained by a *cloze task* on a mass of unlabeled text in an unsupervised way. We run PyTorch[6] framework with CUDA on NVIDIA GTX 3090 GPUs. The learning rates is set to 1e−5, the hypergraph layer to 2, the causal connection threshold for hyperedge construction to 0.7, and all trainable parameters are randomly initialized from normal distributions. We use the Adam optimizer with $L2$ regularization and combine dropout for model training.

5. Results and analysis

5.1. Overall results

Tables 1 and 2 compare the overall performance between our EHNEM and the competitors on the ESC corpus; While Table 3 compares the performance on the English-MECI corpus. The competitors in Tables 1 and 3 have reported all intra-sentence, cross-sentence, and overall results on ESC and English-MECI datasets; While the competitors in Table 2 have only reported the intra-sentence results on the ESC dataset, respectively. Besides, we note that the competitors in Table 1 all exploit some document-level information to encode event mentions for causality classification; While the competitors in Table 2 all use some kind of external knowledge to enhance event causality identification.

The first observation is that our EHNEM achieves significant improvements over all competitors in terms of much higher F1 score for intra-sentence, cross-sentence, and overall causal relation classification in Table 1, even some of these competitors have

[3] https://github.com/nlp-uoregon/meci-dataset
[4] https://github.com/huggingface/transformers
[5] https://github.com/pytorch/fairseq/
[6] pytorch.org

Table 1
Overall results of comparison models for document-level event causality identification on the ESC corpus.

Model	Intra-Sentence			Cross-Sentence			Overall		
	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
ILP (NAACL, 2019)	38.8	52.4	44.6	35.1	48.2	40.6	36.2	49.5	41.9
RichGCN (NAACL, 2021)	49.2	63.0	55.2	39.2	45.7	42.2	42.6	51.3	46.6
GESI (SIGIR, 2022)	–	–	50.3	–	–	49.3	–	–	49.4
ERGO (COLING, 2022)	57.5	72.0	63.9	51.6	43.3	47.1	48.6	53.4	50.9
CHEER (ACL, 2023)	56.9	69.6	62.6	45.2	52.1	48.4	49.7	53.3	51.4
SENDIR (ACL, 2023)	65.8	66.7	66.2	33.0	90.0	48.3	37.8	82.8	51.9
Our EHNEM	63.2	70.4	**66.6**	62.3	59.9	**61.0**	62.6	63.1	**62.8**

Table 2
Intra-sentence results of event causality identification on the ESC corpus.

Model	Precision	Recall	F1 score
KnowDis (COLING, 2020)	39.7%	66.5%	49.7%
KnowMMR (IJCAI, 2020)	41.9%	62.5%	50.1%
CauSeRL (ACL-Findings, 2021)	41.9%	69.0%	52.1%
LearnDA (ACL, 2021)	42.2%	69.8%	52.6%
LSIN (ACL, 2021)	47.9%	58.1%	52.5%
DPJL (COLING, 2022)	65.3%	70.8%	**67.9%**
CF-ECI (ACL-Short, 2023)	47.1%	66.4%	55.1%
ECLEP (ACL-Findings, 2023)	49.3%	68.1%	57.1%
Our EHNEM	63.2%	70.4%	66.6%

Table 3
Overall results of comparison models for document-level event causality identification on the English-MECI corpus.

Model	Intra-Sentence			Cross-Sentence			Overall		
	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
MECI-Base (COLING, 2022)	–	–	–	–	–	–	48.7	59.9	53.7
RichGCN (NAACL, 2021)	62.0	68.8	65.2	8.8	21.9	12.6	52.7	64.8	58.1
DPJL (COLING, 2022)	67.0	81.7	**73.6**	–	–	–	–	–	–
ERGO (COLING, 2022)	60.5	73.1	66.2	8.4	34.4	13.5	47.9	69.8	56.8
Our EHNEM	65.7	80.1	72.2	22.9	12.5	**16.2**	63.9	74.3	**68.7**

used graph-based neural networks to learn a document-wise event representation with a large scale of trainable parameters. The outstanding performance of our EHNEM can be attributed to the exploitation of correlated cause–effect structure in a document for jointly identify multiple causal relations with the same event mention; While these competitors identify causal relations on per event-pair basis.

The second observation is that the improvement of our EHNEM in cross-sentence causality identification (10%+) is much more significant than that of intra-sentence (2%+). This can be attributed to its modeling the interdependence of both intra-sentence and cross-sentence causal relations with the same event mention by a hyperedge, so that the intra-sentence event causality with abundant contextual semantics can facilitate the identification of cross-sentence causal relations. Indeed, an intra-sentence event-pair is often mentioned with an explicit relation description in its contextual semantics, it is not unexpected to perform better than the cross-sentence relation identification.

Table 2 compares the intra-sentence performance between our EHNEM and the competitors that have only reported intra-sentence results. It can be observed that our EHNEM can outperform most intra-sentence competitors, except for a small performance gap with the DPJL. The better performance of DPJL might be attributed to its explicitly using a causal cue word for event causality identification. However, such a cue word does not exist in a cross-sentence event-pair. As such, it cannot be applied in the cross-sentence case.

We reproduce some competitors on the English-MECI dataset in Table 3 for fair comparison, including the state-of-the-art algorithms of overall results in Table 1 (ERGO) and intra-sentence results in Table 2 (DPJL), as well as a most representative graph convolutional network based document-level information encoding algorithm (RichGCN) for event causality identification. The MECI-base is a baseline algorithm provided by (Lai et al., 2022). We can observe similar results with the experiments on the ESC dataset in Tables 1 and 2, that is, our EHNEM has achieved significant performance improvements over all competitors, except for a slight performance gap with the DPJL in intra-sentence event causality identification for its utilization of more prior knowledge.

We can also observe that the MECI-base cannot outperforms the RichGCN, ERGO, and our EHNEM in Table 3. This is not unexpected, as the MECI-base is a baseline model that simply uses a PLM to encode the semantic of two input event mentions for causal relation classification. Besides, it can be observed that the performance of both our EHNEM and the competitors are quite poor in cross-sentence event causality identification. This might be attributed to the fact that the English-MECI corpus contains

Table 4
Ablation results of different hypergraph convolution layers for event causality identification on the ESC corpus.

Model	Intra-Sentence			Cross-Sentence			Overall		
	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)	P (%)	R (%)	F1 (%)
EHNEM (layer = 1)	63.1	68.2	65.5	63.1	57.7	60.2	63.1	61.0	61.9
EHNEM (layer = 2)	63.2	70.4	66.6	62.3	59.9	61.0	62.6	63.1	62.8
EHNEM (layer = 3)	62.8	70.0	66.2	61.2	59.7	60.3	61.7	62.9	62.2
EHNEM (layer = 4)	60.8	70.8	65.4	60.6	60.2	60.3	60.7	63.5	62.0
EHNEM w/o HGNN	70.3	60.5	64.8	68.3	43.2	52.5	69.0	48.6	56.7
EHNEM w GCN	63.0	69.1	65.8	61.8	57.2	59.3	62.3	60.9	61.5
EHNEM full (Ours)	63.2	70.4	**66.6**	62.3	59.9	**61.0**	62.6	63.1	**62.8**

very few instances of cross-sentence causal relations, leading to an insufficient training of neural network models. According to our statistics, only 93 cross-sentence event causal pairs are available for model training, with 36 for validation and 32 for testing.

5.2. Ablations study

Module ablation: To examine the module effectiveness, we design the following ablation studies.

- EHNEM (layer) is our EHNEM model using different hypergraph convolution layers.
- EHNEM w/o HGNN is our EHNEM model without the event-hypergraph learning, which only uses the event semantic encoding outputs as the final prediction.
- EHNEM w GCN is our EHNEM model that uses a graph convolutional network (layer=2) to replace the hypergraph convolution neural network for document causal structure learning.

Table 4 presents results of our ablation study. We can first observe that the EHNEM (layer=1) cannot outperform the others using more hypergraph convolution layers, for its insufficient training with fewer convolution layers and trainable parameters. We can also observe that the EHNEM (layer=2) achieves the best performance and the EHNEM with more convolution layers cannot further improve the performance. This might be explained by the over-smoothing effect of a hypergraph neural network. That is, more convolution layers suggest to involve more farther away nodes into the message passing and aggregation for event representation learning.

The second group of Table 4 presents the results of ablation modules. We observe that neither the EHNEM w/o HGNN nor the EHNEM w GCN can outperform the full EHNEM model. This, however, is not unexpected. As the EHNEM w/o HGNN totally ignores the document causal structure that can be used to learn the interdependence of event causal relations from the whole document; While although the EHNEM w GCN uses a conventional graph convolutional network to learn a kind of document-level event representations, it is unable to learn the document correlated causal-effect structure for it using of simple edges that can only connect two event nodes, as such it cannot benefit from the other causal relation edges. As the EHNEM w GCN also includes the event semantic encoding module and uses a graph convolutional network for document causal structure learning, it achieves comparable performance to the EHNEM full (Ours) model in intra-sentence causality identification. However, the performance gap in inter-sentence (−1.7%) and overall (−1.3%) causality identification remains evident.

Document causal structure: To further investigate the impact of correlated cause–effect structure, we plot the overall performance of our EHNEM against different hyperedge degrees in Fig. 3(a).

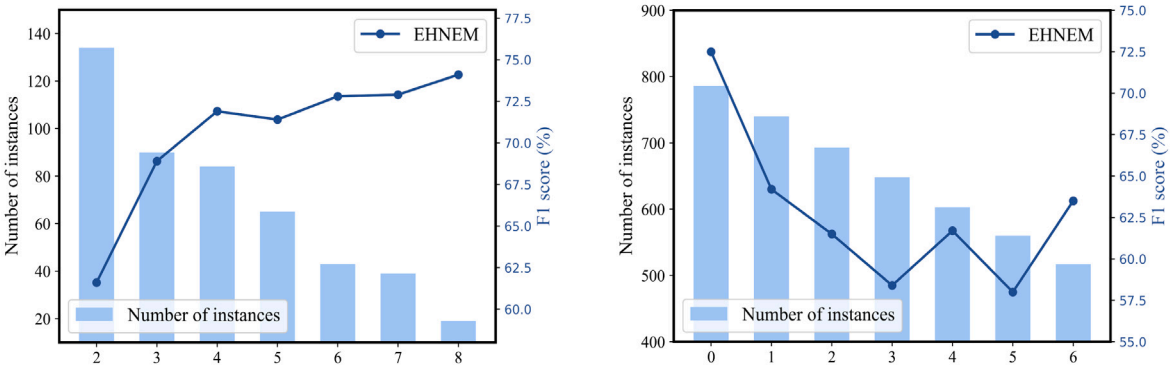
The degree of an event hyperedge represents how many event nodes it connects, that is, the causal relation between its connected nodes are identified jointly. It is not unexpected that our EHNEM benefit from an increase of hyperedge degrees, even the instance number is decreasing. This again validates our argument that learning such correlated cause–effect structure for mining interdependence of multiple cross-sentence events can help improving document-level causal relation identification.

Document-level event causality: To explore the effectiveness of our EHNEM model for document-level causality identification, we plot the overall performance of our EHNEM against different event distances in Fig. 3(b).

We use the number of event mentions between two events as the event interval distance. It is not unexpected that our EHNEM suffers from the increase of event distance, as two farther apart event mentions may contain discontinuous semantics. It is difficult to identify their causal relation mainly from their respective contextual semantics. However, we can observe that its performance does not decrease much with an increase of event distances (the distance of $\{1, \ldots, 6\}$ in the figure). This also validates our argument that constructing a hypergraph-like correlated causal-effect structure can help mining the interdependence among multiple causal relations with the same event for document-level causality identification.

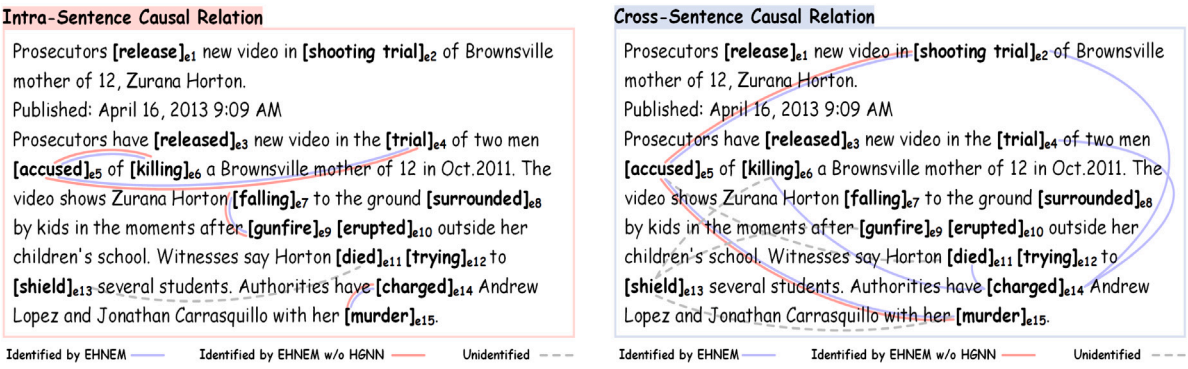
5.3. Case study

We use case study to visualize and compare both intra-sentence and cross-sentence causalities identified by the EHNEM and EHNEM w/o HGNN respectively, as illustrated in Fig. 4. We first observe that both the EHNEM and EHNEM w/o HGNN can identify most intra-sentence causal relation and miss only one event causality pair *(shield; died)*, as illustrated in Fig. 4(a). This



(a) Hyperedge degrees (b) Event distances

Fig. 3. Performance of different hyperedge degrees and different event distances.



(a) Intra-sentence Causal Relation (b) Cross-sentence Causal Relation

Fig. 4. Visualization of the ECI results for both intra-sentence and cross-sentence in a document of the ESC corpus.

is not unexpected. They both contain the event semantic encoding module to encode local event contextual semantics from its belonging raw sentence.

By contrast, for cross-sentence causal relations, the EHNEM w/o HGNN only recognizes two event causality pairs *(shooting trial; accused)* and *(murder; accused)*, but missing many causality pairs that can be recognized by the EHNEM as illustrated in Fig. 4(b). This can be attributed to the document causal structure learning of our EHNEM via a hypergraph convolutional network. We can also observe that those causalities only identified by the EHNEM are with the same effect event "*charged*". This again validates our argument that event causalities in a document are often correlated, which should be exploited to augment document-level causality identification.

5.4. Error analysis

We can first observe that the misidentification of cross-sentence event causality identification is significantly higher compared to intra-sentence identification, despite our proposed EHNEM model has achieved significant improvement in cross-sentence event causality identification (F1 10%+) compared to intra-sentence (F1 2%+), as shown in Table 1. This can be also observed in Fig. 3(b), where the misidentification of intra-sentence causality identification (Distance = 0) is considerably lower than that of cross-sentence (Distance = 1–6). We attribute this misidentification of event causal relations to the semantic discontinuities between events in cross-sentence causal relations; While identifying intra-sentence event causality is relatively easier due to the presence of some explicit relation descriptions within the contextual information.

We can also observe that the misidentification of causal relations without cause–effect correlation (hyperedge degree = 2) is significantly higher compared to that with more cause–effect correlation (hyperedge degree = 4–8), as shown in Fig. 3(a). This again validated our arguments, that is, the identification of one event causality may benefit from other causal relations with the same cause (effect) event. On the other hand, the identification of causal event pair with co-reference events may also benefit from such correlated causal-effect structure. However, the individual event pairs, i.e hyperedge degree = 2, experience more misidentification for the absence of correlated cause–effect events.

6. Conclusion

Existing approaches detect pair-wise causal relation independently, while we argue that the latent correlated cause–effect structure in a document is important for document-level causal relation identification. Motivated from such considerations, we have designed the EHNEM to model latent correlated causal-effect structure by an event-hypergraph and jointly identify multiple causal relations. The EHNEM constructs an event-hypergraph by integrating a potential causal relation for involved event nodes as a hyperedge to capture causal correlations, and a hypergraph convolutional network is used on the constructed event-hypergraph to learn the events' representations for final causality identification. Experiments on the ESC 0.9 corpus have validated our design objective in terms of significant performance improvements over all competitors, with our EHNEM achieving the new state-of-the-art results.

In our future work, we shall investigate how to unequally encode each event pair causal relation according to its importance for document-level overall event causality identification.

CRediT authorship contribution statement

Wei Xiang: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cheng Liu:** Visualization, Methodology, Investigation, Data curation. **Bang Wang:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wei Xiang reports financial support was provided by self-determined research funds of CCNU from the colleges' basic research and operation of MOE and Bang Wang reports financial support was provided by National Natural Science Foundation of China.

Data availability

Data will be made available on request.

References

Baker, C.F., Fillmore, C.J., Lowe, J.B., 1998. The Berkeley FrameNet project. In: 36th Annual Meeting of the Association for Computational Linguistics and 17th International Conference on Computational Linguistics. Association for Computational Linguistics, Quebec, Canada, pp. 86–90, URL https://aclanthology.org/P98-1013/.

Bondarenko, A., Wolska, M., Heindorf, S., Blübaum, L., Ngomo, A.N., Stein, B., Braslavski, P., Hagen, M., Potthast, M., 2022. CausalQA: A benchmark for causal question answering. In: Proceedings of the 29th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, pp. 3296–3308, URL https://aclanthology.org/2022.coling-1.291.

Cao, P., Zuo, X., Chen, Y., Liu, K., Zhao, J., Chen, Y., Peng, W., 2021. Knowledge-enriched event causality identification via latent structure induction networks. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, Virtual, pp. 4862–4872. http://dx.doi.org/10.18653/v1/2021.acl-long.376.

Caselli, T., Vossen, P., 2017. The event StoryLine corpus: A new benchmark for causal and temporal relation extraction. In: Proceedings of the Events and Stories in the News Workshop@ACL 2017. Association for Computational Linguistics, Vancouver, Canada, pp. 77–86. http://dx.doi.org/10.18653/v1/w17-2711.

Chen, M., Cao, Y., Deng, K., Li, M., Wang, K., Shao, J., Zhang, Y., 2022. ERGO: Event relational graph transformer for document-level event causality identification. In: Proceedings of the 29th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, pp. 2118–2128, URL https://aclanthology.org/2022.coling-1.185.

Chen, M., Cao, Y., Zhang, Y., Liu, Z., 2023. CHEER: Centrality-aware high-order event reasoning network for document-level event causality identification. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Toronto, Canada, pp. 10804–10816, URL https://aclanthology.org/2023.acl-long.604.

Fan, C., Liu, D., Qin, L., Zhang, Y., Xu, R., 2022. Towards event-level causal relation identification. In: The 45th International ACM SIGIR Conference on Research and Development in Information Retrieval. Association for Computing Machinery, Madrid, Spain, pp. 1828–1833. http://dx.doi.org/10.1145/3477495.3531758.

Feder, A., Keith, K.A., Manzoor, E., Pryzant, R., Sridhar, D., Wood-Doughty, Z., Eisenstein, J., Grimmer, J., Reichart, R., Roberts, M.E., Stewart, B.M., Veitch, V., Yang, D., 2022. Causal inference in natural language processing: Estimation, prediction, interpretation and beyond. Trans. Assoc. Comput. Linguist. 10, 1138–1158. http://dx.doi.org/10.1162/tacl_a_00511.

Feng, Y., You, H., Zhang, Z., Ji, R., Gao, Y., 2019. Hypergraph neural networks. In: The Thirty-Third AAAI Conference on Artificial Intelligence. AAAI Press, Honolulu, Hawaii, USA, pp. 3558–3565. http://dx.doi.org/10.1609/aaai.v33i01.33013558.

Gao, L., Choubey, P.K., Huang, R., 2019. Modeling document-level causal structures for event causal relation identification. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Association for Computational Linguistics, Minneapolis, MN, USA, pp. 1808–1817. http://dx.doi.org/10.18653/v1/n19-1179.

He, Z., Cao, P., Jin, Z., Chen, Y., Liu, K., Zhang, Z., Sun, M., Zhao, J., 2024. Zero-shot cross-lingual document-level event causality identification with heterogeneous graph contrastive transfer learning. (1), pp. 1–18. http://dx.doi.org/10.48550/arXiv.2403.02893.

Heindorf, S., Scholten, Y., Wachsmuth, H., Ngomo, A.N., Potthast, M., 2020. CauseNet: Towards a causality graph extracted from the web. In: The 29th ACM International Conference on Information and Knowledge Management. ACM, Virtual, pp. 3023–3030. http://dx.doi.org/10.1145/3340531.3412763.

Hu, Z., Li, Z., Jin, X., Bai, L., Guan, S., Guo, J., Cheng, X., 2023. Semantic structure enhanced event causality identification. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Toronto, Canada, pp. 10901–10913, URL https://aclanthology.org/2023.acl-long.610.

Khatib, K.A., Hou, Y., Wachsmuth, H., Jochim, C., Bonin, F., Stein, B., 2020. End-to-end argumentation knowledge graph construction. In: The Thirty-Fourth AAAI Conference on Artificial Intelligence. AAAI Press, New York, NY, USA, pp. 7367–7374, URL https://ojs.aaai.org/index.php/AAAI/article/view/6231.

Lai, V.D., Veyseh, A.P.B., Nguyen, M.V., Dernoncourt, F., Nguyen, T.H., 2022. MECI: a multilingual dataset for event causality identification. In: Proceedings of the 29th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, pp. 2346–2356, URL https://aclanthology.org/2022.coling-1.206.

Liu, J., Chen, Y., Zhao, J., 2020. Knowledge enhanced event causality identification with mention masking generalizations. In: Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence. ijcai.org, Virtual, Japan, pp. 3608–3614. http://dx.doi.org/10.24963/ijcai.2020/499.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V., 2019. RoBERTa: A robustly optimized BERT pretraining approach. pp. 1–13, arXiv preprint URL http://arxiv.org/abs/1907.11692.

Liu, C., Xiang, W., Wang, B., 2024. Identifying while learning for document event causality identification. (1), pp. 1–13. http://dx.doi.org/10.48550/arXiv.2405.20608, arXiv preprint arXiv:2405.20608.

Man, H., Nguyen, C.V., Ngo, N.T., Ngo, L., andThien Huu Nguyen, F.D., 2024. Hierarchical selection of important context for generative event causality identification with optimal transports. In: Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation. ELRA and ICCL, Torino, Italy, pp. 8122–8132, URL https://aclanthology.org/2024.lrec-main.711.

Miller, G.A., 1995. WordNet: A lexical database for English. Commun. ACM 38 (11), 39–41, URL http://doi.acm.org/10.1145/219717.219748.

Mu, F., Li, W., 2023. Enhancing event causality identification with counterfactual reasoning. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Toronto, Canada, pp. 967–975, URL https://aclanthology.org/2023.acl-short.83.

Phu, M.T., Nguyen, T.H., 2021. Graph convolutional networks for event causality identification with rich document-level structures. In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Association for Computational Linguistics, Online, pp. 3480–3490. http://dx.doi.org/10.18653/v1/2021.naacl-main.273.

Pu, R., Li, Y., Wang, S., Li, D., Zheng, J., Liao, J., 2023. Enhancing event causality identification with event causal label and event pair interaction graph. In: Findings of the Association for Computational Linguistics. ACL 2023, Association for Computational Linguistics, Toronto, Canada, pp. 10314–10322, URL https://aclanthology.org/2023.findings-acl.655.

Schuler, K.K., 2006. VerbNet: A Broad-Coverage, Comprehensive Verb Lexicon (Ph.D. thesis). University of Pennsylvania.

Seoh, R., Birle, I., Tak, M., Chang, H., Pinette, B., Hough, A., 2021. Open aspect target sentiment classification with natural language prompts. In: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing. Association for Computational Linguistics, Punta Cana, Dominican Republic, pp. 6311–6322. http://dx.doi.org/10.18653/v1/2021.emnlp-main.509.

Shen, S., Zhou, H., Wu, T., Qi, G., 2022. Event causality identification via derivative prompt joint learning. In: Proceedings of the 29th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, pp. 2288–2299, URL https://aclanthology.org/2022.coling-1.200.

Speer, R., Chin, J., Havasi, C., 2017. ConceptNet 5.5: An open multilingual graph of general knowledge. In: Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence. AAAI Press, San Francisco, California, USA, pp. 4444–4451, URL http://aaai.org/ocs/index.php/AAAI/AAAI17/paper/view/14972.

Sui, Y., Feng, S., Zhang, H., Cao, J., Hu, L., Zhu, N., 2022. Causality-aware enhanced model for multi-hop question answering over knowledge graphs. Knowl.-Based Syst. 250 (1), 108943. http://dx.doi.org/10.1016/j.knosys.2022.108943.

Tao, Z., Jin, Z., Bai, X., Zhao, H., Dou, C., Zhao, Y., Wang, F., Tao, C., 2023. SEAG: Structure-aware event causality generation. In: Findings of the Association for Computational Linguistics. ACL 2023, Association for Computational Linguistics, Toronto, Canada, pp. 4631–4644, URL https://aclanthology.org/2023.findings-acl.283.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Scao, T.L., Gugger, S., Drame, M., Lhoest, Q., Rush, A.M., 2020. Transformers: State-of-the-art natural language processing. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. Association for Computational Linguistics, Online, pp. 38–45. http://dx.doi.org/10.18653/v1/2020.emnlp-demos.6.

Xiang, W., Wang, B., 2019. A survey of event extraction from text. IEEE Access 7 (1), 173111–173137. http://dx.doi.org/10.1109/ACCESS.2019.2956831.

Xiang, W., Wang, Z., Dai, L., Wang, B., 2022. ConnPrompt: Connective-cloze prompt learning for implicit discourse relation recognition. In: Proceedings of the 29th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Gyeongju, Republic of Korea, pp. 902–911, URL https://aclanthology.org/2022.coling-1.75.

Yang, J., Han, S.C., Poon, J., 2022. A survey on extraction of causal relations from natural language text. Knowl. Inf. Syst. 64 (5), 1161–1186. http://dx.doi.org/10.1007/s10115-022-01665-w.

Yao, L., Chu, Z., Li, S., Li, Y., Gao, J., Zhang, A., 2021. A survey on causal inference. ACM Trans. Knowl. Discov. Data 15 (5), http://dx.doi.org/10.1145/3444944.

Yuan, C., Huang, H., Cao, Y., Wen, Y., 2023. Discriminative reasoning with sparse event representation for document-level event-event relation extraction. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. Association for Computational Linguistics, Toronto, Canada, pp. 16222–16234, URL https://aclanthology.org/2023.acl-long.897.

Zhang, B., Chen, Q., Zhou, J., Jin, J., He, L., 2024. Enhancing event causality identification with rationale and structure-aware causal question answering. pp. 1–8. http://dx.doi.org/10.48550/arXiv.2403.11129, arXiv preprint arXiv:2403.11129.

Zhao, K., Ji, D., He, F., Liu, Y., Ren, Y., 2021. Document-level event causality identification via graph inference mechanism. Inform. Sci. 561 (1), 115–129. http://dx.doi.org/10.1016/j.ins.2021.01.078.

Zuo, X., Cao, P., Chen, Y., Liu, K., Zhao, J., Peng, W., Chen, Y., 2021a. Improving event causality identification via self-supervised representation learning on external causal statement. In: Findings of the Association for Computational Linguistics. ACL/IJCNLP 2021, Association for Computational Linguistics, Virtual, pp. 2162–2172. http://dx.doi.org/10.18653/v1/2021.findings-acl.190.

Zuo, X., Cao, P., Chen, Y., Liu, K., Zhao, J., Peng, W., Chen, Y., 2021b. LearnDA: Learnable knowledge-guided data augmentation for event causality identification. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing. Association for Computational Linguistics, Virtual, pp. 3558–3571. http://dx.doi.org/10.18653/v1/2021.acl-long.276.

Zuo, X., Chen, Y., Liu, K., Zhao, J., 2020. KnowDis: Knowledge enhanced data augmentation for event causality detection via distant supervision. In: Proceedings of the 28th International Conference on Computational Linguistics. International Committee on Computational Linguistics, Barcelona, Spain, pp. 1544–1550. http://dx.doi.org/10.18653/v1/2020.coling-main.135.